

CLOUD COMPUTING COURSE

Lecture2-Cloud Computing Characteristic

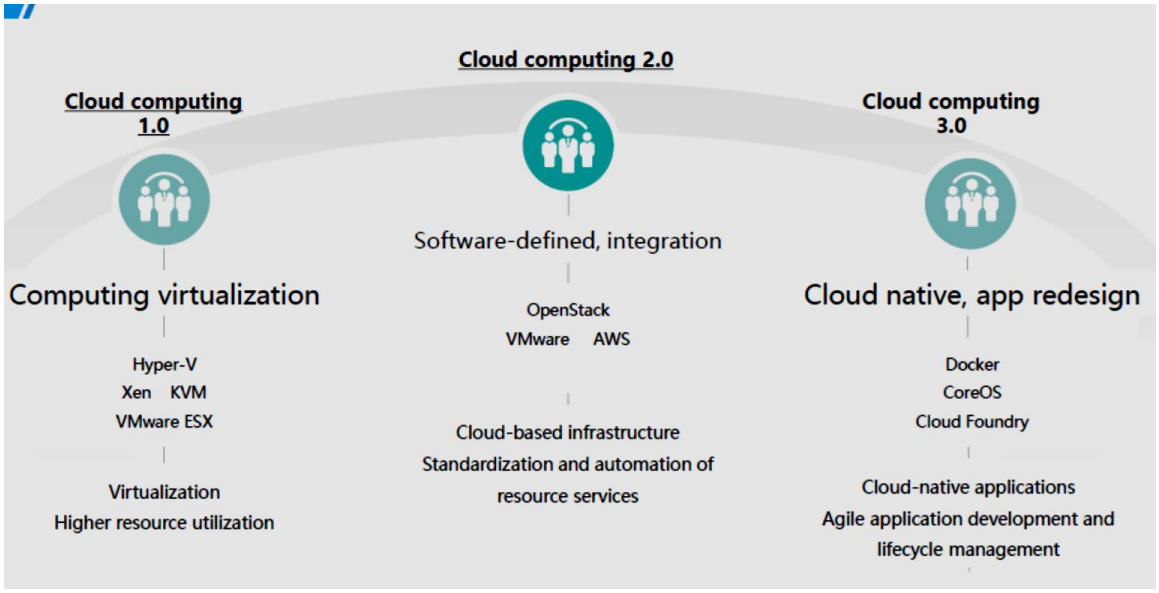
ABSTRACT
Lecture2 introduces cloud-computing characteristic.

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Internet History

ARPANET was born→ **TCP/IP** protocol was established→**DNS** introduced → **WWW** open to public.

Cloud Computing Evolution



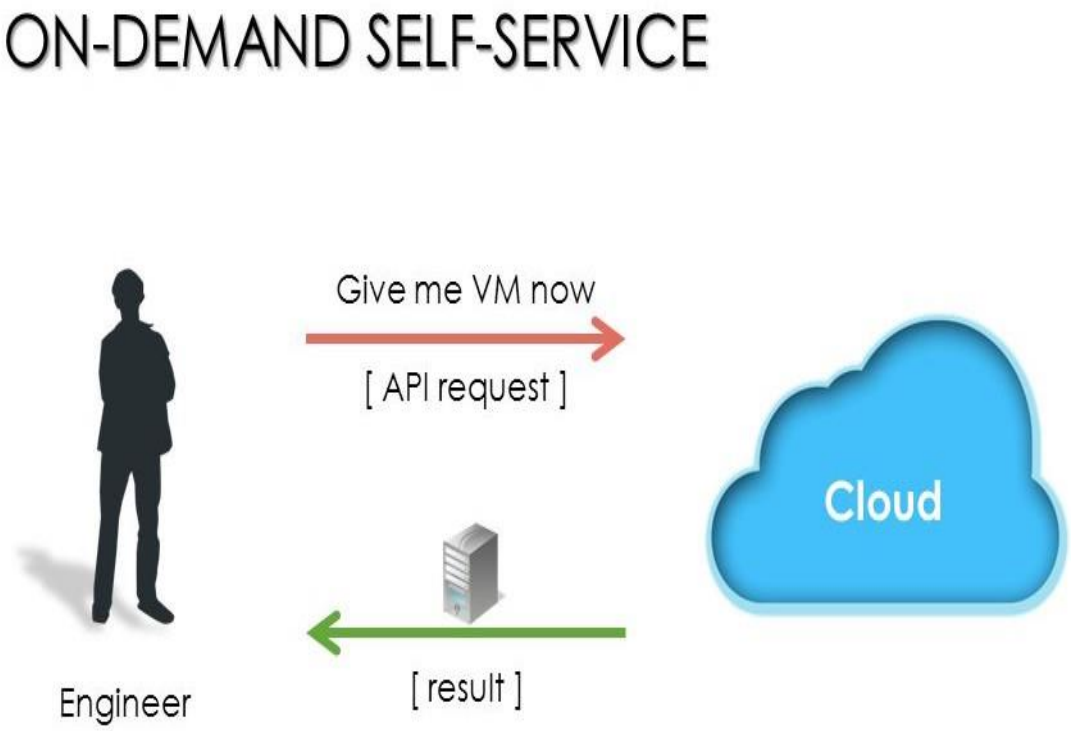
Real Server→Virtual Server→Cloud Virtual Server→ Containers (Applications)→ Serverless Applications

Cloud Computing Characteristic

1. On-Demand Self-Service.

Cloud computing resources can be provisioned without human interaction from the service provider. In other words, a consumer can provision additional computing resources as needed without going through the cloud service provider. This can be a storage space, virtual machine instances, database instances, and so on.

A consumer can use a web self-service portal as an interface to access their cloud accounts to see their cloud services, their usage, and also to provision and de-provision services as they need to.



Perquisites of on-demand self-service:

1. Consumers understanding of their own needs and what product accommodate such needs.
2. Resource Pooling
3. Network connectivity

2. Broad Network Access.

Cloud computing resources are available over the network and can be **accessed by different customer platforms (e.g., mobile phones, tablets, laptops, and workstations).** In other words, cloud services are available over a network such as the internet, or in the case of a private clouds it could be a local area network (LAN).

Network bandwidth and latency are very important aspects of cloud computing and broad network access, because they relate to the quality of service (QoS) on the network. This is particularly important for serving time sensitive manufacturing applications.



3. Resource Pooling.

Resource pooling means that multiple customers are serviced from the same physical resources (**compute, storage and network resources**). Providers' resource pool should be very **large and flexible** enough to service multiple client requirements and to provide for economy of scale. When it comes to resource pooling, resource allocation must not impact performances of critical manufacturing applications.

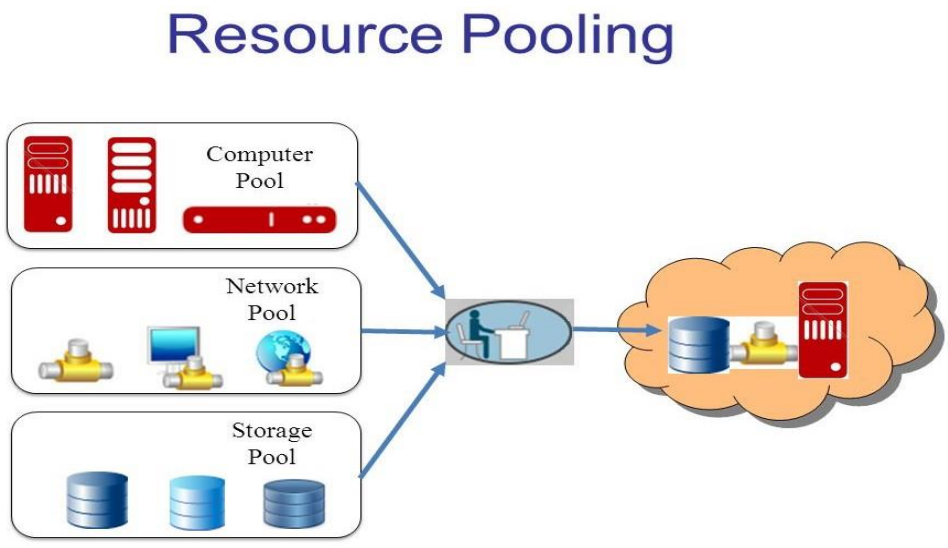
The provider's computing resources are pooled to serve multiple consumers using a multi-tenant model, with different physical and virtual resources dynamically assigned and reassigned according to consumer demand.

Multi-tenancy allows multiple customers to share the same applications or the same physical infrastructure while retaining privacy and security over their information.

The importance of resource pooling:

1. Allow you to have the exact amount of resources you need.
2. Shield the differences in the under lying resources.
3. Provide maximum utilization of the resources.

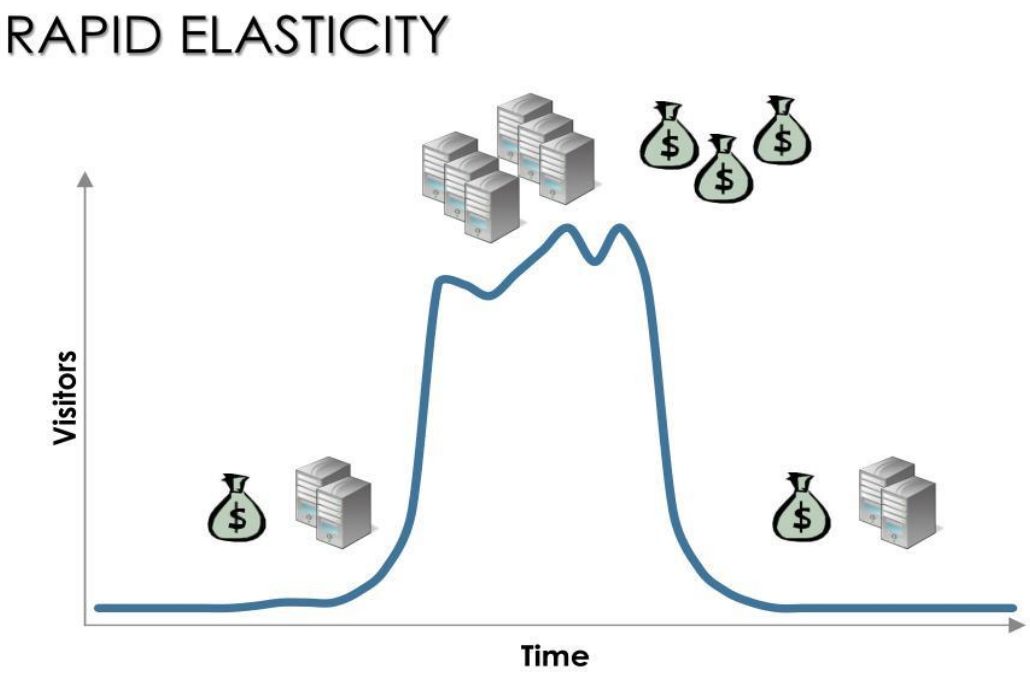
Examples of resources include storage, processing, memory, and network bandwidth.



4. Rapid Elasticity.

One of the great things about cloud computing is the ability to quickly provision resources in the cloud as manufacturing organizations need them. Then to remove them when they do not need them. Cloud computing resources can scale up or down rapidly and, in some cases, automatically, in response to business demands. It is a key feature of cloud computing. The usage, capacity, and therefore cost, can be scaled up or down with no additional contract or penalties.

Elasticity is a landmark of cloud computing and it implies that customers can rapidly provision and de-provision any of the cloud computing resources. Rapid provisioning and de-provisioning might apply to storage or virtual machines or customer applications.



Scaling can be:

Horizontal Scale: this done by increasing number of servers.

Vertical Scale: increasing number of resources within the same server.

5. Measured service.

Cloud computing resources usage is metered and customers pay accordingly for what they have used. Resource utilization can be optimized by leveraging charge-per-use capabilities. This means that cloud resource usage (whether virtual server instances that are running, storage in the cloud, bandwidth, and active user accounts) is monitored, measured and reported by the cloud service provider, providing transparency for both the provider and consumer of the utilized service.

The cost model is based on “pay for what you use”—the payment is variable based on the actual consumption by the customers.



"Pay as you go"

Measuring is based on different criteria’s:

- Duration of usage.

- Resource quota

- Volume of data transmitted.